

2 Impulses are transmitted along nerves as a series of action potentials.

An action potential can be split into four main stages:

- depolarisation
- repolarisation
- hyperpolarisation
- resting state.

(a) (i) Which row shows the events that happen during depolarisation of a neurone?

(1)

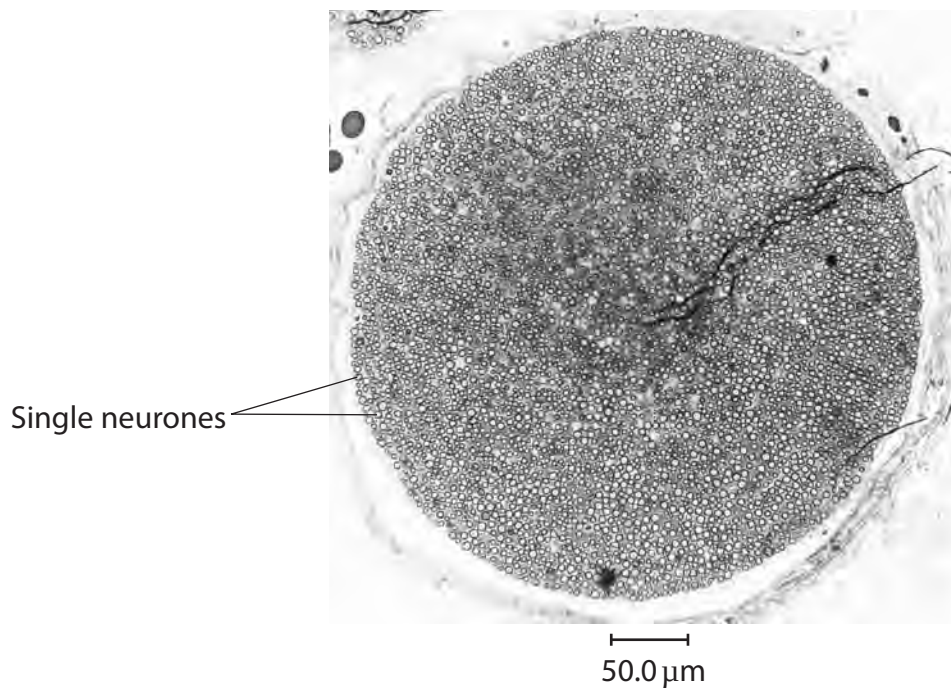
	Sodium channels	Membrane potential
☒ A	closed	decreasing
☒ B	closed	increasing
☒ C	open	decreasing
☒ D	open	increasing

(ii) Explain what happens during repolarisation in a neurone.

(3)

This image shows a full page of white paper with ten horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and extend across the entire width of the page. There is no handwriting or other markings on the paper.

(b) The photograph is a cross-section of part of a sciatic nerve showing the neurones.



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(i) Calculate the magnification of this photograph.

Give your answer in standard form.

(2)

Answer

(ii) Name a type of microscope that would be used to obtain this image.

(1)

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(iii) Explain how myelinated neurones enable a greater speed of transmission in a neurone.

(3)

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(Total for Question 2 = 10 marks)
